

Syeda Shamama Afeef

AI/ML Engineer | Data Scientist

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SUMMARY

AI/ML Engineer with 3.5+ years of experience designing, building, and deploying production-grade AI systems across Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), Machine Learning, and Deep Learning. Experienced in delivering end-to-end AI solutions spanning data engineering, AI/ML model development, backend engineering, MLOps, and cloud deployment across Google Cloud Platform (GCP), Azure, and AWS. Hands-on experience building enterprise AI applications using Google Gemini, LangChain, LangGraph, LlamaIndex, Microsoft Copilot Studio, Azure OpenAI, FastAPI, vector databases, and cloud-native technologies to develop intelligent AI agents, scalable RAG pipelines, and secure AI microservices. Proficient in Python, REST APIs, Docker, Kubernetes, CI/CD, and cloud-native deployments, with a strong focus on enterprise AI, intelligent automation, production-grade LLM applications, and secure, scalable AI solutions for global healthcare and enterprise clients.

EXPERIENCE

AI Engineer, Aviato Consulting

Dec 2025 — Jul 2026

- Collaborated directly with Google Cloud, Google Data Science, and Google AI Engineering teams to deliver the Population Health AI (PHAI) platform for Latrobe Health Service (Australia). Authored the PHAI Implementation Guide, officially adopted by Google and Aviato as the primary project delivery reference.
- Designed and consolidated a modelling-ready healthcare feature store in BigQuery, engineering 471 features by integrating LHS clinical, SISU, and Google PDFM datasets through EDA, statistical analysis, feature engineering, and SQL-based data validation.
- Contributed to the end-to-end Autoencoder → Embedding → K-Means (k=6) pipeline by preparing modelling datasets, validating feature readiness, supporting beam inference, and enabling LLM-driven population health insights across 57 Australian SA4 regions.
- Designed 21 structured system prompts and 9 analytical frameworks for a multi-persona LLM-powered Population Health Insight Engine, incorporating Australian healthcare regulations and domain-specific guardrails to generate reliable, context-aware insights.
- Built BigQuery Conversational AI Agents to review 52+ AI-generated insight tables, achieving 100% factual grounding, and executed secure cross-project BigQuery migrations (52+ tables, master datasets, beam inference, and summary tables) with IAM validation and zero data loss.

Data Scientist (AI/ML), GWEN AI, Aurangabad, India

Mar 2023 — Dec 2025

AI-first startup building voice automation and agentic AI products for enterprise clients.

- Reduced data processing time by **40%** and LLM inference costs by **35%** by architecting production-grade AI systems using **LLMs (GPT-4, Gemini)** and cloud-native AI services for enterprise workflow automation.
- Designed and deployed **Agentic AI systems** and **RAG pipelines** using **LangChain, LangGraph, LlamaIndex, embeddings, and vector databases (Pinecone, ChromaDB)**, enabling autonomous decision making with **~85% accuracy**.
- Built and deployed scalable **ML/DL models, ETL/ELT pipelines, and API-first AI microservices (FastAPI, REST)**, supporting end-to-end AI lifecycle from data ingestion to real-time production inference.
- Implemented end-to-end **MLOps pipelines** using **CI/CD, MLflow, Docker, Kubernetes**, and cloud-native infrastructure, achieving **99.5%+ production uptime**.
- Optimized **LLM accuracy, latency, and cost** through prompt engineering, LLM evaluation, grounding, and intelligent routing while orchestrating workflows with **Apache Airflow**.
- Designed and deployed cloud-native AI applications across Google Cloud Platform (GCP), Azure, and AWS, enabling secure, scalable, and production-ready enterprise AI solutions.

Data Science Intern, BCG, India (Remote)

Dec 2022 — Feb 2023

- **Improved forecast accuracy by 20%** by developing time-series models (LSTM, RNN) using Python (Pandas, NumPy) on financial data
- **Reduced manual reporting effort by 60%** by building automated KPI dashboards (Tableau, Power BI) integrated with SQL pipelines for executive stakeholders
- Delivered **data-driven insights** by collaborating with cross-functional teams, supporting C-suite decision-making

AI Engineer Intern, Cognizant, India (Remote)

Aug 2022 — Nov 2022

- **Reduced model inference latency by 35%** by optimizing ML pipelines using pruning, quantization, and performance tuning techniques
- **Improved workflow efficiency by 25%** by building and deploying NLP classification models (BERT/Transformers, Python) for enterprise automation
- Developed **scalable ML systems and automated reporting workflows** using Python and SQL, reducing manual effort and improving pipeline reliability

Data Analytics Intern, PWC, Switzerland (Remote)

Mar 2022 — Jun 2022

- **Improved predictive accuracy by ~20%** by developing AI-driven risk models using Python and SQL on large-scale financial datasets
- **Reduced reporting time by 40%** by building interactive Tableau dashboards, enabling real-time insights and data-driven decision-making for senior stakeholders

EDUCATION

P.E.S College of Engineering, Aurangabad, India - Bachelor of Technology, Computer Science and Engineering - (GPA: 9.00) Dec 2021 — Jun 2025

CERTIFICATIONS

Generative AI leader, Google	May 2026
Google Advanced Data Analytics Professional, Google	Nov 2023
IBM Data Science, IBM	Jan 2024
Advance Machine Learning, Udemy	Mar 2023
Google Statistics for Data Science, Google	Sep 2023
35+ additional certifications across AI, ML, Cloud, Data Engineering, and DevOps (available on LinkedIn)	Jun 2026

PROJECTS

Self-Healing Agentic MLOps System	
- Reduced manual intervention by ~70% by designing a self-healing ML system with automated failure detection, root cause analysis, and pipeline recovery using multi-agent LLM orchestration (LangChain) - Integrated Evidently AI (data drift) and MLflow (experiment tracking) to trigger automated retraining workflows based on anomaly thresholds - Built real-time observability dashboards (Streamlit) and deployed monitoring infrastructure on AWS, reducing MTTR by ~50% and improving system reliability	
AI Calling Agent (LLM-Powered Voice Automation), Gwen A.I	
- Built an agentic voice automation system for automated interviews, screening, and follow-ups with multi-turn, context-aware conversations - Integrated STT/TTS pipelines with LLM-based intent understanding, enabling natural and low-latency interactions - Designed API-based orchestration with fallback and confidence handling, ensuring reliable real-world deployment	
AI Resume Analyzer (LLM-Powered RAG System), Gwen A.I	
- Reduced manual screening effort by ~65% by building a production-grade resume intelligence platform using LLMs, RAG, and LangGraph-based orchestration - Designed structured extraction pipelines (Pydantic) and LLM-driven summarization workflows with real-time streaming responses (SSE) - Deployed fault-tolerant backend (FastAPI, Docker) with checkpointing and SQLite-based state recovery, ensuring reliable execution at scale	
AI-Driven Credit Card Fraud Detection System, Portfolio	
- Achieved 95%+ precision by developing a fraud detection pipeline using ensemble models (XGBoost, Random Forest) on imbalanced financial datasets - Deployed real-time scoring service using Docker and AWS Lambda with CI/CD, enabling sub-100ms inference at scale - Implemented SHAP-based explainability to deliver interpretable predictions for business stakeholders	

SKILLS

Programming	: Python (Advanced), SQL, JavaScript, TypeScript, Java
Machine Learning & Generative AI	: Scikit-learn, XGBoost, PyTorch, TensorFlow, Deep Learning, Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, LlamaIndex, Prompt Engineering, LLM Evaluation & Fine-Tuning, Embeddings & Semantic Search, Vector Databases (Pinecone, ChromaDB), NLP (Transformers, BERT, spaCy), Computer Vision (YOLOv5, OpenCV), OpenAI APIs, Azure OpenAI, Microsoft Copilot Studio, Google Gemini APIs
Backend Engineering & APIs	: FastAPI, REST APIs, Node.js, Next.js, Microservices, Event-Driven Architecture
Data Engineering & MLOps	: ETL/ELT Pipelines, Feature Engineering, Apache Airflow, CI/CD Pipelines, Docker, Kubernetes, MLflow, Model Validation & Monitoring, LLM Deployment & Observability, Azure DevOps, Git, Bitbucket
Cloud & Platforms	: AWS (SageMaker, Lambda, Athena), Azure (Azure OpenAI, Azure Blob Storage, Azure Neural TTS), Cloud-Native Deployments, GPU Computing (CUDA – Applied), Google Cloud Platform (GCP), Vertex AI, Gemini, BigQuery, Cloud SQL, Azure, Docker, Kubernetes (GKE)
Databases & Analytics	: PostgreSQL, MongoDB, Snowflake, SQLite, Dataverse, Tableau, Power BI, Matplotlib
Compliance & Security	: HITRUST/GRC Compliance, HIPAA-Compliant AI Systems, SSO Integration, Enterprise Security Hardening